# DATA-DRIVEN GRAPH INFERENCE FOR INFRASTRUCTURE NETWORKS

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## MOTIVATION

The objective is to learn a basis that captures the dependencies and global modes of variation (called graph harmonics, or *frequencies*) present in the data. Analyzing how system states (like frequency deviations) decompose into these modes reveals which patterns are most likely to persist or amplify after a disturbance. Thus, given a *meaningful* basis, we can analyze how signals propagate, diffuse, or concentrate, and how oscillations or disturbances (such as power outages) manifest across the entire grid.

## NOTATIONS

### Graph Laplacian

- The Laplacian $\boldsymbol{L} \in \mathbb{R}^{n\times n}$ encodes how values at each node deviate from its neighborhood. We denote the eigendecomposition:

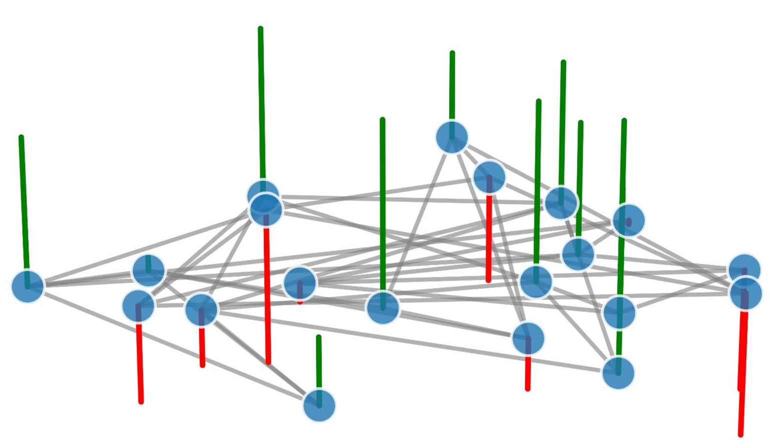
$$\boldsymbol{L} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top,\ \boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$$

- We note the pseudoinverse:

$$\boldsymbol{L}^\dagger = \sum_{k=1}^{n-1} \frac{1}{\lambda_k} \boldsymbol{u}_k \boldsymbol{u}_k^\top$$

### Signal Matrix

- A graph signal is a $n$-valued tensor, collecting states across all nodes.



- Collection $\boldsymbol{X} \in \mathbb{R}^{m\times n}$ of $m$ signals (rows) over $n$ nodes (columns)
- Sample covariance:

$$\boldsymbol{\Sigma} = \frac{1}{m}\boldsymbol{X}^\top \boldsymbol{X}$$

The objective is, given a series of graphs signals, and optional adjacencies, to infer the dependency structure as edge weights.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## LAPLACIAN LEARNING

*$\boldsymbol{L}^\dagger$ encodes graph topology via $\mathbb{E}[\Sigma]$.*

### Edge Interpretation

- $\boldsymbol{\Omega} = \boldsymbol{L}$: Precision matrix
- $\boldsymbol{\Omega}_{i,j} = -W_{i,j}$ (partial correlations)
- $\boldsymbol{\Omega}_{i,i} = D_i$ (degree)

### Conditional Independence

$$x_i \perp x_j \mid \boldsymbol{X}_{\setminus\{i,j\}} \iff \boldsymbol{L}_{i,j} = 0$$

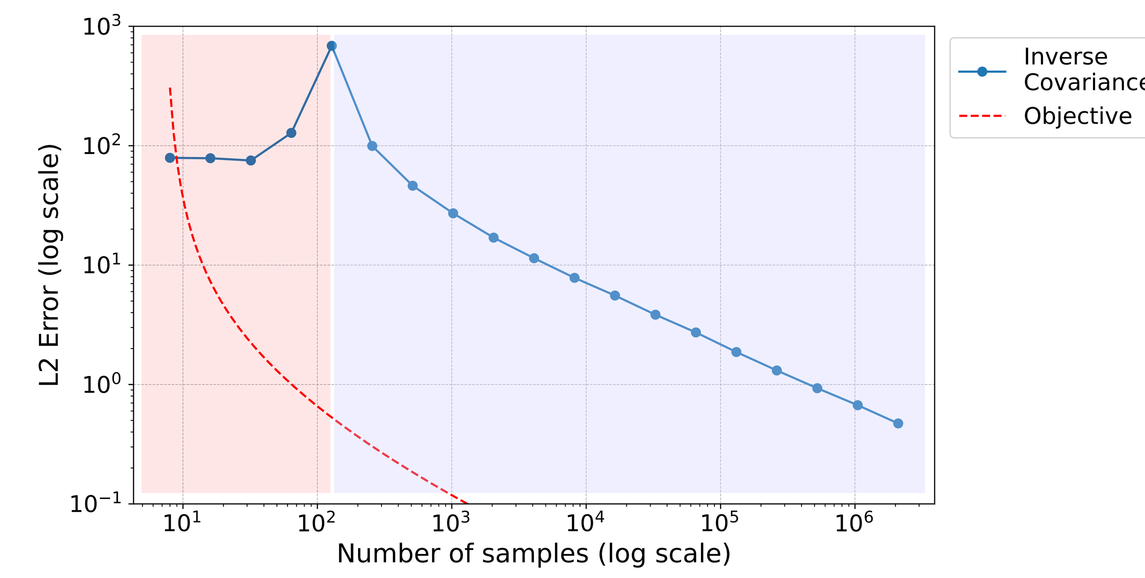
### Maximum Likelihood

$$\max_{\boldsymbol{L}\in\mathcal{L}} \overbrace{\log\det(\boldsymbol{L})}^{\text{Promotes large eigenvalues (smooth surrogate for rank)}} - \overbrace{\operatorname{tr}(\boldsymbol{L}\boldsymbol{\Sigma})}^{\text{Penalizes deviation from covariance}}$$

$$\mathcal{L} = \{\boldsymbol{L} : \underbrace{\boldsymbol{L} \succeq 0}_{\boldsymbol{L}\text{ is positive semidefinite}},\ \underbrace{\boldsymbol{L}\boldsymbol{1} = \boldsymbol{0}}_{\text{Rows sum to zero (centering)}},\ \underbrace{\boldsymbol{L}_{i\neq j} \leq 0}_{\text{Off-diagonal entries are non-positive}}\}$$

The determinant of a rank deficient (square) matrix is always zero, which happens:

1. When $\boldsymbol{\Sigma}$ is rank-deficient ($k < n$), as we minimize $\operatorname{tr}(\boldsymbol{L}\boldsymbol{\Sigma})$.
2. When enforcing Laplacian constraints (hence singularity by design).




### Sparsity Penalty

$$\min_{\Theta\succ 0} \left\{ \operatorname{tr}(\Theta\Sigma) - \log|\Theta| + \beta \sum_{i\neq j} \rho(\Theta_{ij}) \right\}$$

- The sparse penalty function $\rho(\cdot)$ (e.g. $\ell_1$, SCAD, MCP or LCP) prevents edge weight shrinkage and overfitting.
- It also helps with stability by preventing $\boldsymbol{\Theta}$ from becoming nearly singular.

### Ridge Regularization

- We learn the shifted precision $\boldsymbol{\Theta} + \delta\boldsymbol{I} \succ 0$ under the constraint $\boldsymbol{\Theta} \in \mathcal{L}$.

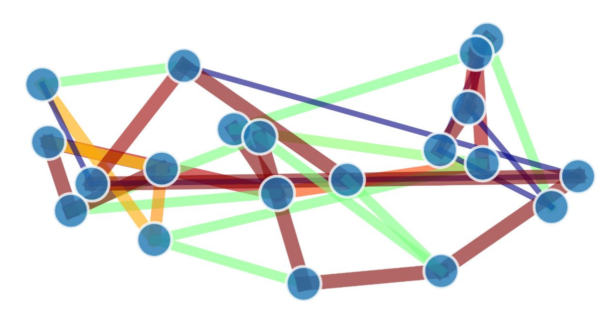
### Spectral Relaxation

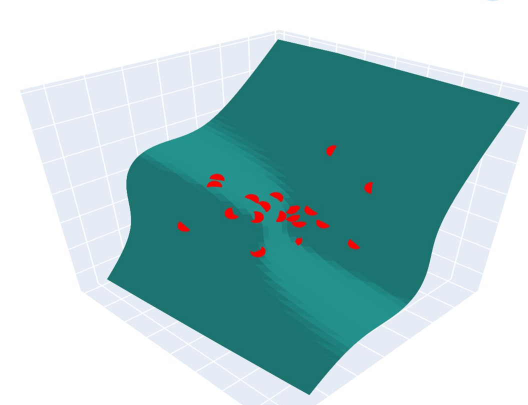
$$\min_{\boldsymbol{L}\in\mathcal{L}} \left\{ \operatorname{tr}(\boldsymbol{L}\boldsymbol{\Sigma}) + \alpha_1 \|\boldsymbol{L}\|_{F,\text{off}}^2 - \alpha_2 \sum_i \log(\boldsymbol{L}_{ii}) \right\}$$

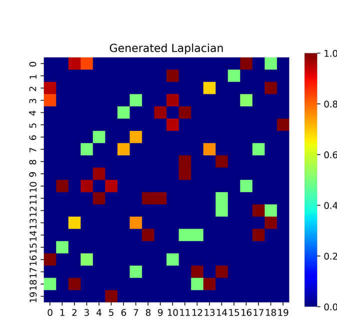
- Strongly convex due to Frobenius norm, while implicitly enforcing spectral properties.
- Stabilizes estimation when $\Sigma$ is rank-deficient.

## MANIFOLD STRUCTURE

- As $n \to \infty$ and bandwidth $\epsilon \to 0$, $\boldsymbol{L} \to \Delta$ (Laplace-Beltrami) under uniform sampling.








- Dirichlet energy, whose minimization solves $\Delta f = 0$:

$$\mathcal{E}(f) = \int_{\mathcal{M}} \|\nabla f\|^2 \leftrightarrow \boldsymbol{x}^\top \boldsymbol{L}\boldsymbol{x}$$

- The Laplacian eigenvalues and eigenvectors provide a Fourier-like basis for graph signals, enabling frequency analysis.

The task shifts from learning a graph Laplacian that reflects data covariance, to identifying low-dimensional manifolds that preserve local neighborhoods.

- Diffusion maps learn a set of orthonormal basis functions—the eigenvectors of the diffusion operator $H_t$:

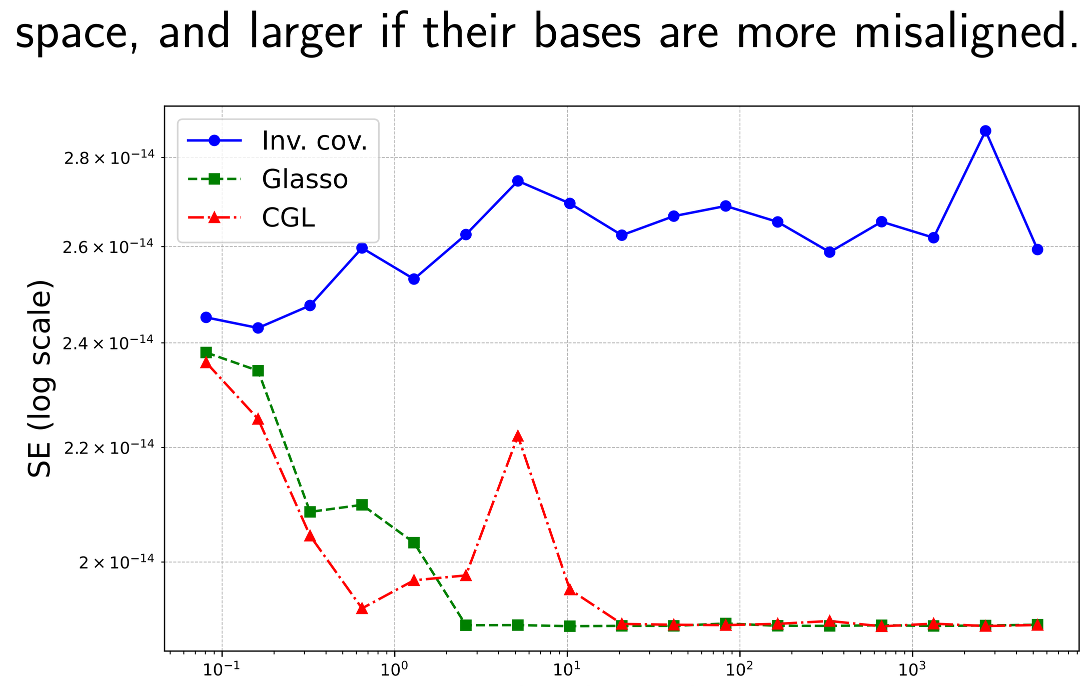
$$\frac{\partial \boldsymbol{f}}{\partial t} = -\boldsymbol{L}\boldsymbol{f} \iff \boldsymbol{f}(t) = \underbrace{e^{-t\boldsymbol{L}}}_{H_t} \boldsymbol{f}_0$$

## ALIGNMENT RESULTS

Given two symmetric matrices $\boldsymbol{L}, \hat{\boldsymbol{L}}$ and their projection matrices $\boldsymbol{P}, \hat{\boldsymbol{P}}$, the subspace error is:

$$\text{SE} = \|\boldsymbol{P} - \hat{\boldsymbol{P}}\|_F, \quad \boldsymbol{P} = \boldsymbol{U}\boldsymbol{U}^\top$$

This measures how different the two subspaces (spanned by the bases $\boldsymbol{U}, \hat{\boldsymbol{U}}$ in which the Laplacians are diagonal) are: it is zero if they span the same subspace, and larger if their bases are more misaligned.




## FUTURE WORK

### Interference and Non-Linearity

- Can we generalize diffusion-based approaches to distortion-based ones?
- Can we go beyond i.i.d. time series?
- Can we have theoretical guarantees for nonlinear dynamics?

### Application to Infrastructures

- Water flow in water distribution systems (WDSs): Can a data-driven Laplacian help producing signals that are compressible or smooth?
- Water quality and propagation in sensor networks (over WDSs): How can we infer contaminant transport from partially observed signals?
- Power grid stability: How can we capture ways in which disturbances propagate across a grid?